Patient Monitoring System (PMS)

Biomedical Engineering — Assignment Breakdown

Design a 5-bed centralised patient monitoring system with nurse station control, digital health integration, and hospital management linkage.

OVERALL GOALS

G1 **Physical ward design**
Define a safe, functional 5-bed layout with instrumented bedside units

G2 **Centralised monitoring**
All beds feed live vitals to a single nurse station hub

G3 **Digital health integration**
Abha ID registration + MongoDB-backed record sync

G4 **Data modelling**
Full RDBMS schema covering patients, vitals, alerts, and staff

G5 **HMS linkage**
Plug PMS into hospital-wide admission, billing, and inventory

MILESTONES

Phase 1 — Research & planning

M1 — Understand requirements

Document

- List all 5-bed ward requirements from the brief
- Note VOC instruments: diathermy, pulse oximetry, patient safety
- Identify relevant safety standards (IEC 60601)

Deliverable: Requirements checklist (1 page)

M2 — Research instruments & standards

Document

- Spec each bedside device: ECG, NIBP, SpO2, temperature
- Note power ratings, signal types, isolation class
- Research Abha ID API and De-health/FHIR standards

Deliverable: Instrument specification table

Phase 2 — Design

M3 — Ward layout & electrode diagram

Design

- Draw floor plan: 5 bed positions, nurse station, cable routing
- Show electrode lead placement per bed (12-lead or 3-lead ECG)
- Mark patient safety zones and grounding points

Deliverable: Ward layout schematic + electrode diagram

M4 — System architecture diagram

Design

- Block diagram: Bed unit -> data bus -> CPU hub -> nurse station
- Show alarm signal path and acknowledgement loop
- Add Abha ID portal and MongoDB sync connections

Deliverable: System block diagram (labelled)

M5 — RDBMS schema design

Design

- Define tables: Patients, Beds, Instruments, Vitals_Log, Alerts, Staff
- Set foreign keys, indexes (bed_id + timestamp), constraints
- Draw ER diagram

Deliverable: ER diagram + schema DDL

Phase 3 — Integration design

M6 — Digital health flow

Integration

- Map Abha ID registration -> patient record creation flow
- Design MongoDB sync schedule (real-time vs periodic)
- Define FHIR resource mapping for vitals and summaries

Deliverable: Data flow diagram

M7 — HMS integration design

Integration

- Define admission/discharge trigger hooks into HMS
- Map billing events: device usage, consumable tracking
- Sketch inventory alert system for electrodes, probes

Deliverable: HMS integration map

Phase 4 — Write-up & submission

M8 — Draft all sections

Write-up

- Introduction, ward layout, instrument spec, architecture
- RDBMS schema section with ER diagram embedded
- Digital health + HMS integration sections
- Safety & standards discussion (IEC 60601, alarm design)

Deliverable: Full draft document (8 sections)

M9 — Review & finalise

Write-up

- Cross-check all diagrams are referenced in text
- Verify instrument specs against standards cited
- Proofread and format for submission

Deliverable: Final submission-ready document